

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain piezoelectric method of ultrasonic wave generation. (CO5)

Q.20 i) Explain Molecular Theory of Surface Tension. (CO2)

ii) Convert force of 200 Newton into dynes. (CO1)

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Roll No.

**Level 3, 1st Sem / DVOC (Ref. & Air Cond.,
Medical Imaging Tech., Auto. Servicing, ITM, PT,
SD, AMT, FP, EMS**

Subject : Applied Physics

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 The S.I unit of Electric Charge is: (CO1)

- a) Ampere b) Coulomb
- c) Farad d) None of these

Q.2 The Friction in flowing fluid is called: (CO2)

- a) Viscosity b) Surface Tension
- c) Density d) None of these

Q.3 The Frequency of a body in free vibrations is called: (CO3)

- a) Free Frequency b) Own Frequency
- c) Natural frequency d) None of them

Q.4 SI unit of Heat is:- (CO4)

- a) Joule b) Kelvin
c) Calorie d) None of these

Q.5 Very High Frequency sound waves are called _____ (CO5)

- a) Infrasonic wave b) Ultrasonic Wave
c) Both (a) & (b) d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Define Reflection. (CO6)

Q.7 Define fundamental unit. (CO1)

Q.8 Give an example of forced vibration. (CO3)

Q.9 Give an use of Overhead Projector. (CO6)

Q.10 S.I unit of Temperature. (CO4)

(2)

188413

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Write down the dimensional formula of Area, Volume, Acceleration, Force and Torque. (CO1)

Q.12 Define surface tension and explain its molecular theory. (CO2)

Q.13 A vibrating system consists of a mass of 200kg and spring constant is 80N/mm. Find the natural frequency of the vibration of this system. (CO3)

Q.14 State the Laws of refraction. (CO6)

Q.15 Write down the properties of heat radiation. (CO4)

Q.16 Define ultrasonic waves. Explain any two application of ultrasonic waves in industry. (CO5)

Q.17 Write down the difference between Heat and Temperature. (CO4)

Q.18 Briefly Describe the concept of Capillary Action. (CO2)

(3)

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